

Extremal equal-spin Einstein–Gauss–Bonnet black holes at $T = 0$: reproduction of arXiv:2303.12471 and the extremal mass shift

October 7, 2026

1 What was done

1. The equal-spin, asymptotically flat part of arXiv:2303.12471 (Kleihaus, Kunz, Radu; KKR below) was reproduced: the near-horizon closed forms of its section 3.2 and the extremal boundary of the domain of existence in the left panels of its fig. 1, against curves extracted from the vector EPS files of the arXiv source.
2. A spectral solver was built for the *extremal* solutions themselves, in KKR’s gauge, with the two zero modes of that gauge identified and fixed. It gives $M_{\text{ext}}(J, \alpha)$, $\mu(y) = M_{\text{ext}}/J^{2/3}$ and the slope $\partial M_{\text{ext}}/\partial \alpha|_J = \mu'(y)$ directly at $T = 0$, the latter by implicit differentiation of the discrete system.
3. The result was compared with the manuscript’s $T \rightarrow 0$ extrapolation of non-extremal solutions (`results/egb-extremality.json`).
4. The perturbations of the extremal throat were analysed: the horizon is not smooth at finite coupling.

Units: $r_H = 1$, i.e. $g_H = 1$ for the round part of the horizon metric, as in the manuscript; J is the angular momentum in *each* plane; $y = \alpha/J^{2/3}$, $x = 3\pi\alpha/(4M)$, $j = (3/2)^{3/2}\sqrt{\pi}J/M^{3/2}$.

2 Field equations and checks

The reduced Lagrangian of arXiv:1010.0860 (as transcribed and validated in the project with $g = r^2$) was used with the squashing function g left free, since the extremal gauge needs it. Its Euler–Lagrange equations for (b, f, g, h, w) agree with $-\sqrt{bh/f} g (G + \alpha H)^{\mu\nu} \partial g_{\mu\nu} / \partial v$ from the independent tensor evaluator on generic, non-solution jets of all five functions (`derive.py`).

3 The near-horizon geometry and a non-smooth horizon

On $b = v_1 \rho^2$, $g^{\rho\rho} = \rho^2/v_1$, $g = G_0$, $h = H_0$, $w' = w_1$ the equations reduce to three algebraic relations. Their solution reproduces KKR’s eqs. (at7)–(at9) to round-off on 60 points of their v_3 parametrisation, with one convention difference: the “ J ” of their eq. (at9) is $J_1 + J_2$, twice the per-plane J of their asymptotic formula. At $\alpha = 0$ it is the extremal Myers–Perry throat, $S = 2\pi J$.

Linearising about the throat with $\delta \propto \rho^\gamma$ (gauge $g^{\rho\rho}$ fixed) gives exponents in AdS_2 pairs $(\gamma, -1 - \gamma)$. At $\alpha = 0$ they are the integers $\{-3, -2, 0, 0, 1, 2\}$; at finite coupling the growing exponent 2 becomes a non-integer $\gamma(y)$ (fig. 1). It falls to 1.773 at $y = 0.029$, returns through 2 at $y = 0.71$ and through 3 at $y = 4.51$ (scanned up to $y = 36.3$). A generic extremal solution excites this mode, so for $\gamma < 2$ its metric functions are C^1 but not C^2 in ρ at the horizon (a statement about the metric functions; curvature in a parallelly propagated frame was not computed). This

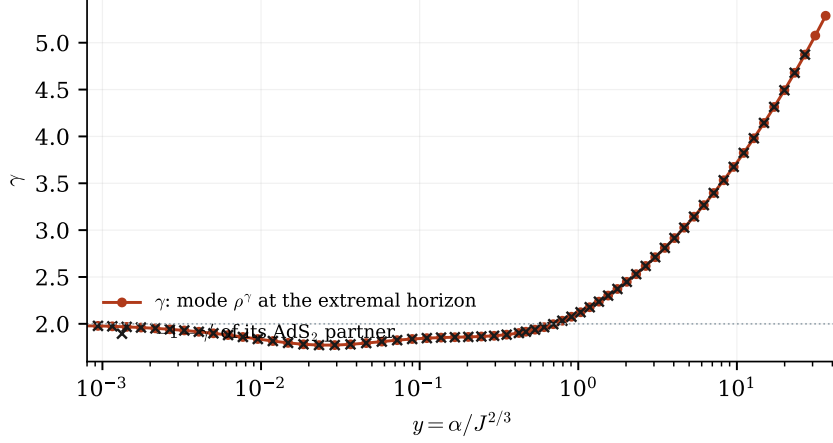


Figure 1: Exponent of the growing perturbation of the extremal throat, $\delta \propto \rho^\gamma$, and of its AdS_2 partner $-1 - \gamma'$; integer ($\gamma = 2$) only at $y = 0$ and at isolated couplings.

is, as far as recalled and not re-read here, the equal-spin Gauss–Bonnet counterpart of the non-smooth extremal horizons reported for Kerr with higher-derivative corrections (arXiv:2303.07358). It also means that KKR’s “approximate form of the solutions as a power series in r ” at the horizon (their $r^2 \propto \rho$) cannot be exact: the expansion needs $r^{2\gamma}$ terms.

How this was checked. The exponents come from the linear analysis, which is integer at $\alpha = 0$ and pairs correctly at every coupling. In the full solutions the effect shows up indirectly: in a representation where the mass is read from a second derivative at infinity, it converges only algebraically, while an analytic solution converges geometrically; and the constraint violation and the resolution differences peak next to the horizon. A direct fit of the Chebyshev tail of the production solutions does *not* resolve the exponent, because the tail reaches the round-off plateau by $n \approx 38$, before an algebraic decay $n^{-(4\gamma+1)}$ becomes visible.

4 The extremal solver

Gauge. KKR’s ansatz with $a = 1$ and $P_i = e^{2F_i}$: $b = P_1 r^4 / (u^2 + 1)$, $g^{rr} = P_1^{-1} r^2 / u$, $g = P_2 u$, $h = P_2 P_3 u(1 + u^{-2})$, $w = \sqrt{2}/(u^2 + 1) + W$, $u = r^2 + 1$. The product $b g^{rr}$ is fixed. This is what makes the extremal problem well posed numerically. In the gauge $g = r^2$, with the double zero factored out of b and f , nearby non-extremal solutions are almost representable by polynomials (both b and f pick up the same $1/x$ factor), and the collocation matrix became singular as N grew. That route was abandoned.

Zero modes. With a and α fixed the gauge still leaves two exact zero modes, identified from smooth, resolution-independent null vectors of the Jacobian: (i) the physical size of the black hole relative to a , and (ii) the residual radial gauge $r \rightarrow r + c\sqrt{b g^{rr}}$, which is $\propto r^3$ at the horizon and a constant shift at infinity. They are fixed by $g_H = P_2(0) = 1$ and by the absence of a $1/r$ term in P_2 .

Discretisation. Chebyshev collocation in $s = r/(1 + r)$, in which the residual-gauge term r^3 and the asymptotic tails are smooth and only $r^{2\gamma}$ remains singular. The unknowns are Q_i with $P_i = 1 + (1 - s)^2 Q_i$ and $W = (1 - s)^2 Q_4$. This builds in asymptotic flatness and the gauge condition, and makes the mass a value at infinity, $M = \frac{3\pi}{4}(1 - c_t)$ with $c_t = Q_1(1)/2$. In the P representation c_t was $P_1''(1)/4$, weighted by k^4 in the Chebyshev tail and converging only algebraically. J is the conserved momentum of w , $J = -\frac{\pi}{16} \partial L / \partial w'$, evaluated in the interior.

At $s = 0$ the E_{P_i} rows vanish identically and are replaced by $dP_i/ds = 0$; at $s = 1$ all four equations vanish identically in this representation and are collocated at a point between the last two nodes. The residuals were generated symbolically in three expanded forms (near each end and in between) to avoid cancellation. The Jacobian is assembled by the chain rule and agrees with a dense complex-step Jacobian to round-off.

Validation. At $\alpha = 0$ Newton returns extremal Myers–Perry with $M = 3\pi/4$, $J = \pi\sqrt{2}/4$, $\Omega_H = 1/\sqrt{2}$, $S = 2\pi J$, $j = 1$, $a_H = s = \frac{1}{2}$. Over the branch the Hamiltonian constraint, which is not imposed, stays below 7.3×10^{-5} . The horizon data agree with the near-horizon solution at the same α/g_H : $P_1(0)/4 = v_1$ to 3.3×10^{-6} , $h_H = H_0$ to 2.4×10^{-7} . Two identities are evaluated, not imposed. Over the 183 branch points whose two resolutions agree to 10^{-5} in μ and μ' : the first law at $T = 0$ along the branch, $\Psi = M_\alpha - 2\Omega_H J_\alpha$, agrees with the implicitly differentiated slope to 3.7×10^{-6} (median 7.2×10^{-8}); Smarr at $T = 0$, $\Psi = (M - 3\Omega_H J)/\alpha$, to 2.8×10^{-5} (median 5.4×10^{-7}) for $\alpha \geq 0.05$, since it divides by α ; and the identity $3\omega = \mu - y\mu'$ of the manuscript holds to 5.2×10^{-5} . The acceptance threshold on the constraint was 10^{-6} up to $\alpha \approx 0.76$, 10^{-5} up to $\alpha \approx 2.29$ and 10^{-4} beyond; the comparison with the manuscript (table 1) uses separate high-resolution solutions, not the branch. Spurious roots of the discrete system exist (non-decaying Chebyshev tails, constraint $\sim 10^{-3}$) and are rejected by those two diagnostics. At larger coupling Newton needs a tangent predictor to stay on the physical root.

5 Reproduction of fig. 1 of arXiv:2303.12471

The vector paths of the six `*-MP2.eps` panels were mapped to data coordinates through each plot’s own tick labels. Against the closed-form static curves the calibration is good to 3.7×10^{-4} , about one device unit. On the range we construct ($x \leq 0.863$), our extremal branch agrees with KKR’s extremal curve to 6.5×10^{-4} in $a_H(x)$ and 1.4×10^{-3} in $s(x)$, and to 1.2×10^{-3} and 2.6×10^{-3} in the insets against j (fig. 2). The boundaries $j \leq 1$ and $x < 1$ hold; j falls to 0.189 at the largest coupling reached. KKR continue to $x \approx 0.92$ and draw a hand extrapolation to the singular static point $(x, j) = (1, 0)$. Our branch stops at $\alpha = 3.118$ ($y = 2.443$, $x = 0.863$): beyond it the constraint violation, concentrated next to the horizon, exceeds 10^{-4} even at $N = 312$, and the round-off floor of the residual rules out higher N . Going further needs a coordinate that resolves the horizon region better, or extended precision, not more continuation.

6 The extremal mass shift

Fig. 3 shows $\mu(y)$ and $\mu'(y)$ on the branch. The slope starts at π , falls to 1.0710 at $y = 0.195$ and rises again, reaching 2.1775 at the end of the branch; μ grows by 203% over the range. Table 1 compares the extremal solutions, computed at $N = 64, 96, 128$ at each of the manuscript’s 12 finite couplings, with the manuscript’s $T \rightarrow 0$ extrapolations at the same α . Across them y agrees to 3.4×10^{-7} and μ to 6.2×10^{-6} . The manuscript’s Ψ_{ext} agrees with the directly computed slope to 1.9×10^{-5} , at most 1.8×10^{-3} of its own fit-sensitivity envelope. The extrapolated central values are therefore far more accurate than the envelope suggests. Where the two differ in μ by more than the manuscript’s three-fit spread, the difference is still at the 10^{-6} level.

7 What was not checked

- No independent check of the field equations beyond the tensor route on generic jets and the closed forms; the near-horizon identification is KKR’s.
- The resolution differences are differences, not error bounds. The round-off floor of the residual grows with N and limits the attainable accuracy to about 10^{-7} – 10^{-6} in μ ; above

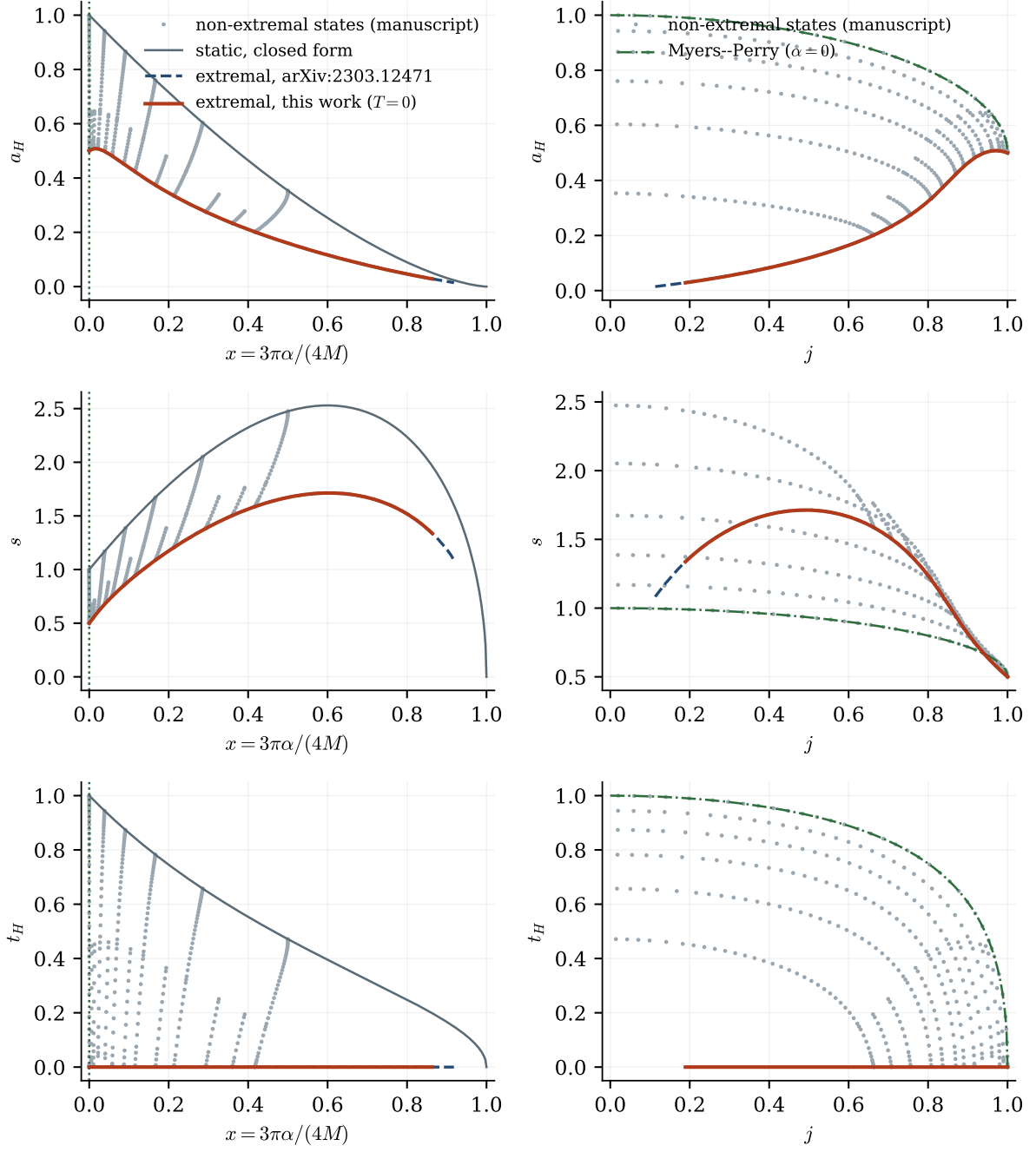


Figure 2: Left panels of fig. 1 of arXiv:2303.12471. Left column against $x = 3\pi\alpha/(4M)$, right column (their insets) against j . Grey: non-extremal states of the manuscript; solid grey-blue: static family (closed form); dash-dotted green: Myers-Perry; dashed blue: KKR's extremal curves, from their EPS; solid red: extremal solutions constructed here at $T = 0$.

α	y	μ (this work)	res.	μ (ms)	spread	μ' (this work)	res.	Ψ_{ext} (ms)	envelope
0.005	0.004790	2.2115781	9.9×10^{-7}	2.2115792	1.3×10^{-3}	2.990940	5.4×10^{-6}	2.990943	1.1×10^{-1}
0.01	0.009845	2.2262777	7.1×10^{-7}	2.2262788	7.7×10^{-4}	2.824561	3.5×10^{-6}	2.824558	8.8×10^{-2}
0.02	0.020720	2.2550577	1.2×10^{-6}	2.2550579	1.6×10^{-3}	2.472846	4.4×10^{-6}	2.472827	1.6×10^{-1}
0.035	0.038427	2.2944442	9.4×10^{-7}	2.2944380	8.4×10^{-4}	2.000655	2.0×10^{-6}	2.000660	9.0×10^{-2}
0.05	0.056889	2.3280166	8.0×10^{-7}	2.3280172	5.7×10^{-6}	1.659079	9.0×10^{-7}	1.659080	8.6×10^{-3}
0.075	0.087750	2.3734822	2.2×10^{-7}	2.3734823	9.1×10^{-6}	1.325970	9.2×10^{-8}	1.325970	2.4×10^{-3}
0.1	0.117873	2.4107272	1.5×10^{-6}	2.4107287	3.7×10^{-5}	1.166215	1.9×10^{-8}	1.166214	8.3×10^{-4}
0.15	0.175201	2.4739927	4.7×10^{-7}	2.4739932	1.1×10^{-5}	1.071493	1.3×10^{-7}	1.071492	2.3×10^{-3}
0.2	0.229060	2.5319360	4.3×10^{-7}	2.5319363	8.8×10^{-6}	1.090071	1.5×10^{-7}	1.090070	2.8×10^{-3}
0.3	0.329212	2.6465810	1.2×10^{-6}	2.6465821	4.5×10^{-6}	1.207815	4.1×10^{-7}	1.207813	2.1×10^{-3}
0.4	0.422511	2.7651786	1.2×10^{-6}	2.7651797	4.6×10^{-6}	1.333843	4.3×10^{-7}	1.333842	2.7×10^{-3}
0.5	0.511244	2.8885340	8.0×10^{-7}	2.8885349	1.1×10^{-6}	1.444398	3.5×10^{-7}	1.444398	6.2×10^{-4}

Table 1: Extremal solutions at $T = 0$ against the manuscript’s $T \rightarrow 0$ extrapolation (ms). “res.” is the larger of the differences between $N = 128$ and 96 and between 96 and 64: a resolution difference, not a bound. “spread” is the manuscript’s three-fit degree spread of μ , “envelope” its twenty-fit sensitivity of Ψ_{ext} .

$N \approx 100$ more resolution does not help.

- The non-smooth exponent is established by the linear analysis; the full solutions are consistent with it but do not measure it independently.
- The branch does not reach $x \approx 0.92$ as KKR do; the endpoint conjecture $(x, j) \rightarrow (1, 0)$ is not tested.
- The digitised curves of KKR are compared on the range we construct; no statement is made about their accuracy beyond the figure.
- The interior of the domain of existence is shown only through the manuscript’s states ($\alpha \leq \frac{1}{2}$); KKR’s interior scan was not repeated.

8 Run receipt

- date of this build: 2026-10-07; directory `work/kkr-extremal`
- environment: Python 3.10.12, NumPy 2.2.6, SciPy 1.15.3, SymPy 1.14.0 (project `.venv`)
- `eps_digitise.py`; `kkr_curves.py` $\rightarrow$ `digitised.json`, `kkr_curves.json`
- `derive.py` $\rightarrow$ `equations.pkl` (tensor check)
- `nearhorizon.py` $\rightarrow$ `nearhorizon.json` (closed forms, exponents)
- `kkr_equations.py` 0; `pw_generator.py` 0 $\rightarrow$ generated residuals and J
- `test_rq.py` $\rightarrow$ `convergence_rq.json` (gate, resolution study)
- `branch_rq.py` (two phases) $\rightarrow$ `branch_rq.json`
- `polish_couplings.py` $\rightarrow$ `polished.json`
- `analyse_branch.py`; `gamma_check.py` $\rightarrow$ `analysis.json`, `gamma_check.json`
- `marimo export html figures_kkr.py` $\rightarrow$ `figures/`, `figures_kkr.html`
- `make_report_numbers.py` $\rightarrow$ `report/numbers.tex`, `receipt.tex`

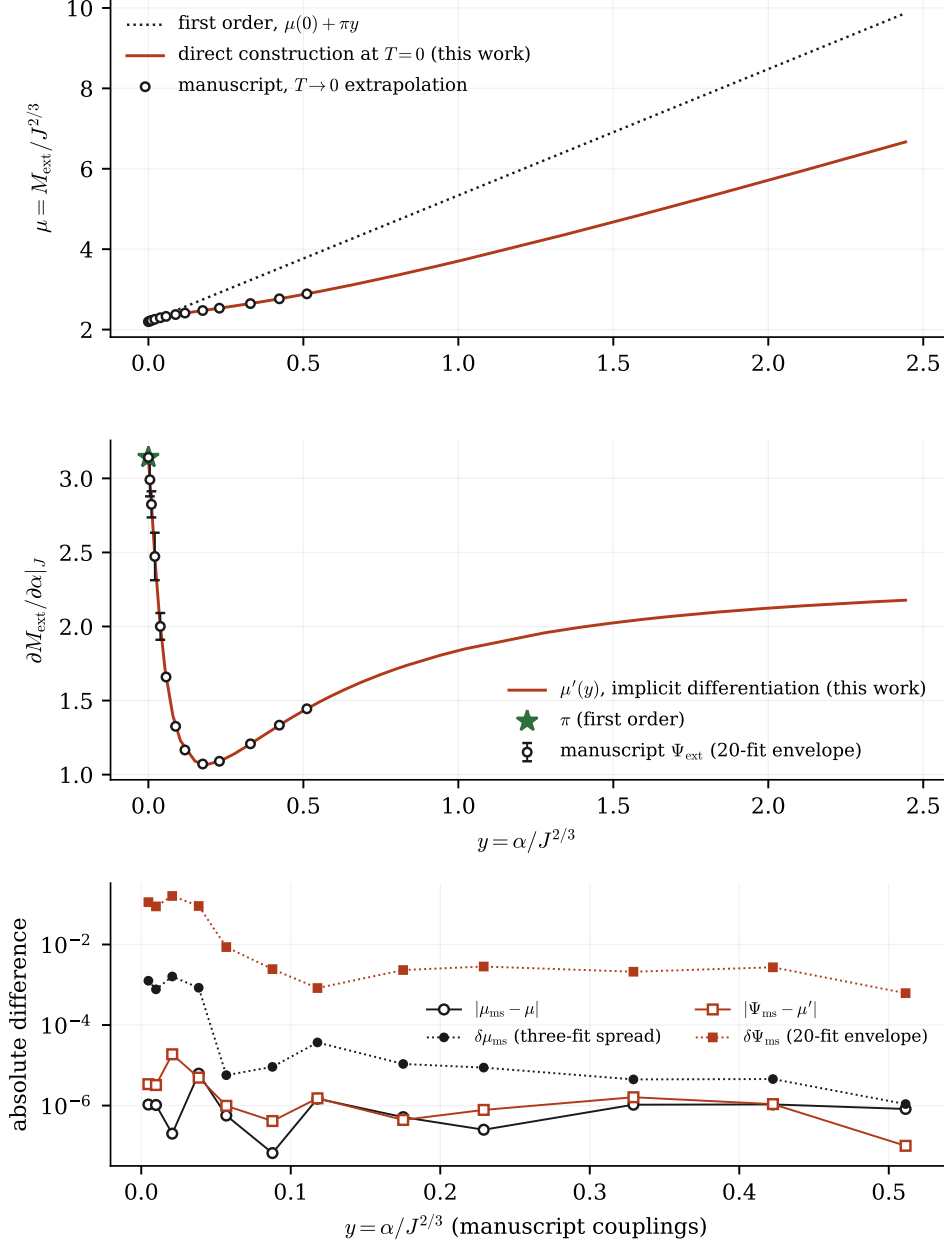


Figure 3: Top: $\mu(y)$ from the extremal solutions (line) and the manuscript's extrapolations (circles); dotted: first order. Middle: slope by implicit differentiation (line) and the manuscript's Ψ_{ext} with its envelope. Bottom: differences in units of the manuscript's own spreads.